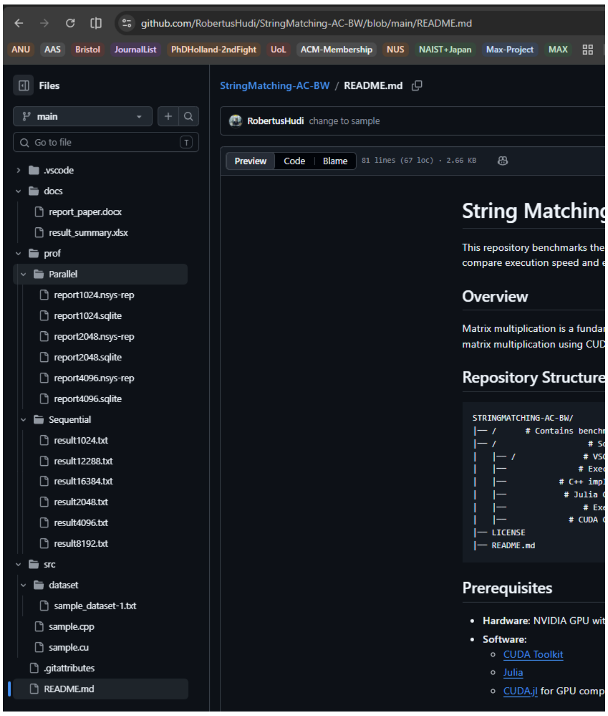
Parallel Computing Final Project 2026

GPU Computing using CUDA

General Guidelines

1. Students are expected to choose one of the given topics (see moodle Session 6) to work on.
2. Required files and documents are necessary to be submitted as zipped file and must be present on your team repository in Github:
   a. Source Files: sequential code (.cpp), parallel code (.cu), dataset (in-folder) and all header or automation file (.sh, .hpp, .c, .py, etc. – if any),
   b. Profiling Results: NVIDIA Nsight System files (.nsys-rep and .sqlite),
   c. Summary Document: Profiling Result Summary (.xlsx and/or .py) and Research Paper Report (.doc or .docx)
   d. Readme file.

   Following is an example of how your repository should look like:



3. Final submission of the zipped repository file will be done in 2 deadlines: Monday, 27 April 2026 for **draft version**, and Monday, 4 May 2026 for the **final version**.
4. Citation guidelines for Research Paper Report:
   a. Papers listed on reference list section should be taken from reputable journal or proceedings with following rules:
      - Minimum reference used is 18 papers;
      - At least 12 of the papers listed on the Reference List **has to be** cited on the text.
      - Main reference is given alongside the topic list on the moodle, **must be** used.
      - First 10 references (including main reference) on the Reference List has to be taken from journals or proceedings with publication date from year 2021-2026.
      - For the rest of the paper on the reference list, try your best to take reference later than year 2010, unless the paper is used for fundamental concept reason.

b. Style of the citation used on any part of your text should be following instructions on the template, please pay attention to this rule.
c. References appearance on any part of your text should be in exact same order as it appears on the reference list. (for example, in IEEE format, paper with number [1] on reference list should appear first in the text, and so on.)
d. Reference paper with same first author's name should not be used for more than 3 titles.

Research Paper Report Template

Choose one of these paper format to write your final report:

- IEEE Paper Format Report. (Find the format here: IEEE Paper Template)
- Springer Nature Paper Format Report (Find the format here: Springer Paper Template, use splnproc2510.docm inside the zipped file)

Research Guide

1. [C/C++] Find the code and dataset for one of the algorithm on following links:
   a. GPU-Accelerated Sliding-Window Feature Extraction for Constraint-Aware DNA Data Storage
   b. Constraint-Aware PFAC Motif Matching on GPUs for High-Throughput DNA Storage Streams
   c. Unified GPU Fusion Kernel for Window-Based Constraint Metrics and Motif Signals in DNA Storage Systems
   d. GPU-Accelerated LSBM Algorithm for DNA-Storage Encoding Scheme

   Dataset Sample could be obtained based on providence of each main paper given, you may find them on authors github repositories as well.

2. Instruction to compile and run the code file could be found in point 4 and 5.

```
student@hudi-Z790-UD-AX:~/string_matching/KMP-on-CUDA$ ls
a.out  direct_match.cpp  direct_match.cu  dm  input.txt  kmp.cpp  kmp_CUDA.cu  output.txt  README.md
student@hudi-Z790-UD-AX:~/string_matching/KMP-on-CUDA$ nvcc kmp_CUDA.cu -o kmp_CUDA
student@hudi-Z790-UD-AX:~/string_matching/KMP-on-CUDA$ ./kmp_CUDA input.txt
95 2
----Start copying data to GPU----
----Data copied to GPU successfully---- Takes 0.000000 seconds
----String matching done---- Takes 0.000078 s
----Task done---- Takes 0.000000 seconds in total
student@hudi-Z790-UD-AX:~/string_matching/KMP-on-CUDA$
```

3. Modify the dataset to fit the input format for each code file. Dataset per-processing is important to differentiate the pattern and data banks as these datasets will be measured by its size. **List of the sequences** is recommended to be **generated manually** from a function thus the length and variation can be controlled. Refer to this generator function in case you need it:

```
// generate random sequence of length n
void seq_gen(int n, char seq[]) {
  for (int i = 0; i < n; i++) {
    int base = rand() % 4;
    switch (base) {
    case 0:
      seq[i] = 'A';
      break;
    case 1:
      seq[i] = 'T';
      break;
```

```
    case 2:
      seq[i] = 'C';
      break;
    case 3:
      seq[i] = 'G';
      break;
    }
  }
}
```

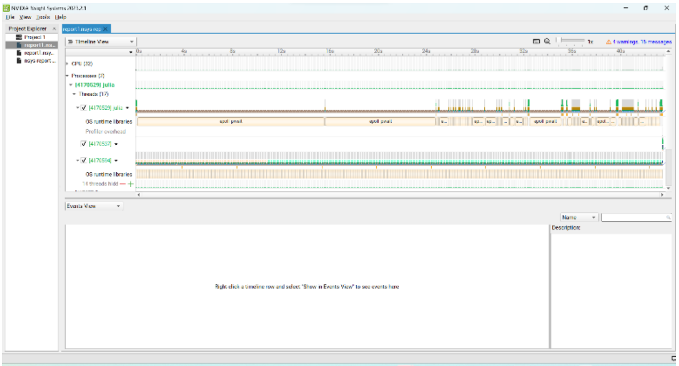
4. Make sure the execution is tracked and labelled with the correct sequence size so that you will not lose track of the profiling later. Check on the algorithm correctness as well.
   (Sample Execution Command) -> `nvcc kmp_8192.cu -o kmp_8192`
   `./kmp_8192 10 >> result_kmp_8192_10.txt`

5. Conduct profiling with this command (make sure you are in the same directory as your code file):
   [C/C++]

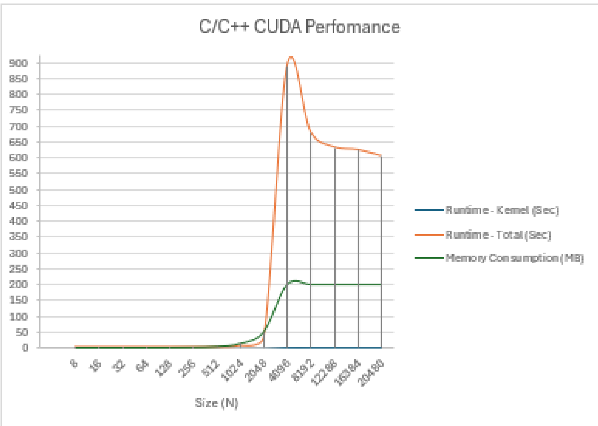
```
nsys profile  -o report --stats=true ./<program_name>.exe
```

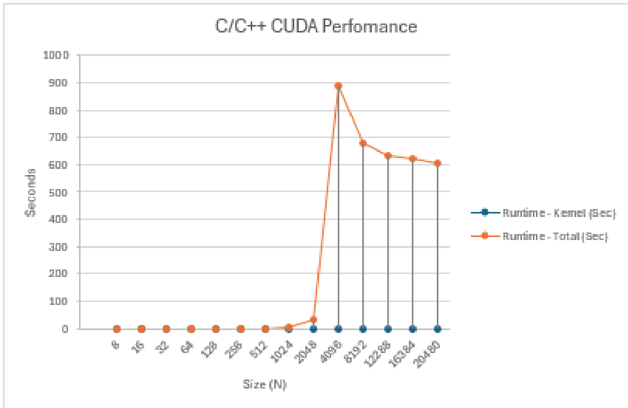
   Which generates files: report.nsys-rep, change the filename if needed.

6. Open the *nsys-rep* files using NVIDA Nsight Systems (use the version according to your CUDA Toolkit), it should look like this:



7. Take the data of execution time and memory consumption data of each kernel, and for the whole process of the application, put it into the table then convert the table into graph for a better analysis.
   <u>Table sample</u>:





Size (N)	Runtime - Kernel (Sec)
8	0.00000272
16	0.000003904
32	0.000006336
64	0.000011392
128	0.000037728
256	0.000078433
512	0.00057853
1024	0.004018382
2048	0.031514411
4096	0.257320488
8192	2.058869684
12288	30.82358426

Size (N)	Runtime - Total (Sec)
8	8.458
16	8.79
32	8.87
64	8.936
128	8.965
256	8.905
512	8.786
1024	8.82
2048	8.711
4096	10.05
8192	18.81
12288	55.91

Size (N)	Memory Consumption (MB)
8	0.003
16	0.008
32	0.033
64	0.131
128	0.524
256	2.097
512	8.388
1024	33.555
2048	134.217
4096	536.871
8192	2147.484
12288	4831.839

8. Following are guidelines for each chapter title for the final report:
   - *Abstract*
   - Chapter I – Introduction
   - Chapter II – Problem Statement and Potential Solution
   - Chapter III – Methodology and Experimental Setup
   - Chapter IV – Result and Analysis
   - Chapter IV – Conclusion